A Counterexample to the Convergence of Three-Block ADMM with an Identity Third Constraint Block*

Kenan Xu†, Xiangfeng Wang‡, Bingsheng He§

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Abstract

The alternating direction method of multipliers (ADMM), as a landmark algorithm, has attracted tremendous research attention and extensive practical applications over the past two decades [1]. It is well known that, although the two-block ADMM enjoys well-established theoretical convergence guarantees, its direct extension to the three-block case may fail to converge, as demonstrated by existing counterexamples [3]. However, for a special case in which the coefficient matrix of the third block variable in the linear equality constraint is the identity matrix, the convergence of the directly extended ADMM remains an open problem: neither a rigorous theoretical proof nor a counterexample is currently available. In this paper, we aim to answer this open problem about ADMM. It is shown that the direct extension of three-block ADMM may nevertheless fail even when the first two blocks are strongly convex quadratics. We can formally construct an explicit counterexample with a strongly convex objective function and linear constraints satisfying the required conditions, such that when ADMM is directly applied to solve this problem, the resulting sequence fails to converge (forming a bounded nonconvergent sequence of minimal period 66). Furthermore, a small perturbation to the dual-variable update suffices to break the non-convergent cycle and yield a convergent method. The counterexample and further analysis were discovered in an AI-assisted workflow using Codex with the GPT-5.6 Sol model.

1 The Open Problem

Consider the linearly constrained three-block convex program

$$\begin{aligned} \min_{x_1,x_2,x_3} \quad & \theta_1(x_1)+\theta_2(x_2)+\theta_3(x_3) \\ \text{s.t.} \quad & A_1x_1+A_2x_2+A_3x_3=b, \end{aligned} \tag{1.1}$$

where, for $i=1,2,3$, $x_i\in\mathbb{R}^{n_i}$, $A_i\in\mathbb{R}^{m\times n_i}$, $b\in\mathbb{R}^m$, and $\theta_i:\mathbb{R}^{n_i}\to(-\infty,+\infty]$ is proper, closed, and convex. Block constraints may be absorbed into θ_i through indicator functions, and the solution set is assumed nonempty. Although two-block ADMM is convergent under standard assumptions [1, 4], its direct three-block extension can fail [3]. We study the subclass $n_3=m$, $A_3=I_m$.

*Code and certificates will be archived with the final version.

†School of Mathematical Sciences, East China Normal University, Shanghai 200241, P.R. China.

‡Key Laboratory of Mathematics and Engineering Applications (MoE) and School of Mathematical Sciences, East China Normal University, Shanghai 200241, P.R. China.

§School of Mathematics, Nanjing University, Nanjing 210093, P.R. China

This structure arises from the inequality-constrained problem

$$\begin{aligned} \min_{x,y} \quad & F(x) + G(y) \\ \text{s.t.} \quad & Ax + By \le b, \end{aligned} \tag{1.2}$$

where $F : \mathbb{R}^{n_x} \to (-\infty, +\infty]$ and $G : \mathbb{R}^{n_y} \to (-\infty, +\infty]$ are proper, closed, and convex, $A \in \mathbb{R}^{m\times n_x}$, $B \in \mathbb{R}^{m\times n_y}$, and $b \in \mathbb{R}^m$. All vector inequalities are understood componentwise, and $\mathbb{R}^m_+ := \{u \in \mathbb{R}^m : u \ge 0\}$. For a nonempty closed convex set C, let ι_C denote its indicator: $\iota_C(u) = 0$ for $u \in C$ and $\iota_C(u) = +\infty$ otherwise. Introducing a nonnegative slack variable gives

$$\begin{aligned} \min_{x,y,z} \quad & F(x) + G(y) + \iota_{\mathbb{R}^m_+}(z) \\ \text{s.t.} \quad & Ax + By + z = b. \end{aligned} \tag{1.3}$$

This is (1.1) with $(x_1, x_2, x_3) = (x, y, z)$ and $(A_1, A_2, A_3) = (A, B, I_m)$; the last subproblem is projection onto $\mathbb{R}^m_+$. We consider the direct slack-last sweep, rather than a grouped two-block reformulation [8].

For $\beta > 0$ and $\lambda \in \mathbb{R}^m$, we use the augmented Lagrangian

$$\mathcal{L}_\beta(x, y, z, \lambda) = F(x) + G(y) + \iota_{\mathbb{R}^m_+}(z) - \lambda^\top r + \frac{\beta}{2}\|r\|^2, \qquad r = Ax + By + z - b \in \mathbb{R}^m. \tag{1.4}$$

Assuming that the subproblems attain solutions, direct three-block ADMM in the order $x \to y \to z \to \lambda$ is

$$x^{k+1} \in \arg\min_x \mathcal{L}_\beta(x, y^k, z^k, \lambda^k), \tag{1.5a}$$

$$y^{k+1} \in \arg\min_y \mathcal{L}_\beta(x^{k+1}, y, z^k, \lambda^k), \tag{1.5b}$$

$$z^{k+1} \in \arg\min_z \mathcal{L}_\beta(x^{k+1}, y^{k+1}, z, \lambda^k), \tag{1.5c}$$

$$\lambda^{k+1} = \lambda^k - \beta(Ax^{k+1} + By^{k+1} + z^{k+1} - b). \tag{1.5d}$$

We use the normal-cone convention

$$N_C(u) = \{v : \langle v, w - u\rangle \le 0 \text{ for every } w \in C\}, \qquad u \in C.$$

With the sign convention in (1.4), a KKT point satisfies

$$\begin{gathered} A^\top\lambda^\star \in \partial F(x^\star), \qquad B^\top\lambda^\star \in \partial G(y^\star), \\ Ax^\star + By^\star + z^\star = b, \qquad z^\star \in \mathbb{R}^m_+, \qquad \lambda^\star \in N_{\mathbb{R}^m_+}(z^\star). \end{gathered} \tag{1.6}$$

Thus $\lambda^\star \le 0$ and $(z^\star)^\top\lambda^\star = 0$.

Available multi-block convergence results impose additional regularity or parameter assumptions [12, 2, 10, 11], or modify the iteration by correction, proximal, or back-substitution steps [5, 9, 7, 6]. These guarantees do not follow from $A_3 = I_m$ alone. Section 2 gives a rational $A = B = I_2$ instance with a bounded non-KKT ADMM sequence of minimal period 66.

2 The counterexample

This section proceeds from instance to orbit to mechanism to certificate. We state the rational QP and the period-66 result, visualize the cycle, then use the projection active sets to obtain a piecewise-affine map and prove exact closure, minimality, and non-KKT status.

2.1 Counterexample data and main result

Set $A = B = I_2$, $\beta = 1$, and consider

$$\begin{aligned} \min_{x,y,z\in\mathbb{R}^2} \quad & \frac{1}{2}x^\top Q_1 x + \frac{1}{2}y^\top Q_2 y + \iota_{\mathbb{R}^2_+}(z) \\ \text{s.t.} \quad & x + y + z = \bar{b}, \end{aligned} \tag{2.1}$$

Define

$$M = (Q_1 + I_2)^{-1}, \qquad N = (Q_2 + I_2)^{-1}.$$

Then (1.5) becomes

$$x^{k+1} = M(\bar{b} - y^k - z^k + \lambda^k), \tag{2.2a}$$
$$y^{k+1} = N(\bar{b} - x^{k+1} - z^k + \lambda^k), \tag{2.2b}$$
$$z^{k+1} = \left[\bar{b} - x^{k+1} - y^{k+1} + \lambda^k\right]_+, \tag{2.2c}$$
$$\lambda^{k+1} = \lambda^k - (x^{k+1} + y^{k+1} + z^{k+1} - \bar{b}). \tag{2.2d}$$

Set

$$\varepsilon = \frac{1}{1000}, \qquad \mu = \frac{8957}{10000}, \qquad \nu = \frac{999}{1000}, \qquad d_1 = (-1, 20)^\top, \qquad d_2 = (-1, 10)^\top, \tag{2.3}$$

and define

$$\begin{aligned} M &= \varepsilon I_2 + (\mu - \varepsilon)\frac{d_1 d_1^\top}{d_1^\top d_1}, \quad Q_1 = M^{-1} - I_2, \\ N &= \varepsilon I_2 + (\nu - \varepsilon)\frac{d_2 d_2^\top}{d_2^\top d_2}, \quad Q_2 = N^{-1} - I_2. \end{aligned} \tag{2.4}$$

The fixed vectors d_1, d_2 determine the eigendirections of the two quadratic Hessians.

To define the right-hand side, prescribe a KKT point. Choose

$$z^\star = (0, 1)^\top, \qquad \lambda^\star = (-1, 0)^\top,$$

and set

$$x^\star = Q_1^{-1}\lambda^\star, \qquad y^\star = Q_2^{-1}\lambda^\star, \qquad \bar{b} = x^\star + y^\star + z^\star. \tag{2.5}$$

By construction,

$$Q_1 x^\star = \lambda^\star, \qquad Q_2 y^\star = \lambda^\star, \qquad x^\star + y^\star + z^\star = \bar{b}, \qquad \lambda^\star \in N_{\mathbb{R}^2_+}(z^\star),$$

so $(x^\star, y^\star, z^\star, \lambda^\star)$ satisfies the KKT conditions (1.6). Every entry of $Q_1, Q_2, \bar{b}$ is rational. Moreover,

$$\sigma(Q_1) = \left\{999, \frac{1043}{8957}\right\}, \qquad \sigma(Q_2) = \left\{999, \frac{1}{999}\right\}, \tag{2.6}$$

where $\sigma(Q)$ denotes the spectrum. Thus $Q_1, Q_2 \succ 0$. Strong convexity in (x, y) and the constraint determine a unique primal solution, while $Q_1 x^\star = \lambda^\star$ fixes the multiplier. Hence the KKT point is unique. All three ADMM subproblems also have unique minimizers, so the iteration is single-valued.

Theorem 2.1 (Exact rational period-66 counterexample)**.** *For the rational problem* (2.1)–(2.5)*, there is a rational initialization for which the sequence generated by the unmodified direct three-block ADMM* (1.5) *is bounded, non-KKT, and periodic with minimal period* 66*. Its strict projection-sign sequence is*

$$\mathcal{W} = (00)^2(01)^{64}, \tag{2.7}$$

where the ith bit is 1 *if the corresponding projection input is positive and* 0 *if it is negative. All* 132 *projection inequalities are strict with a common margin greater than* 10^{-3}*. Therefore an identity third constraint block does not guarantee unconditional global convergence of direct three-block ADMM.*

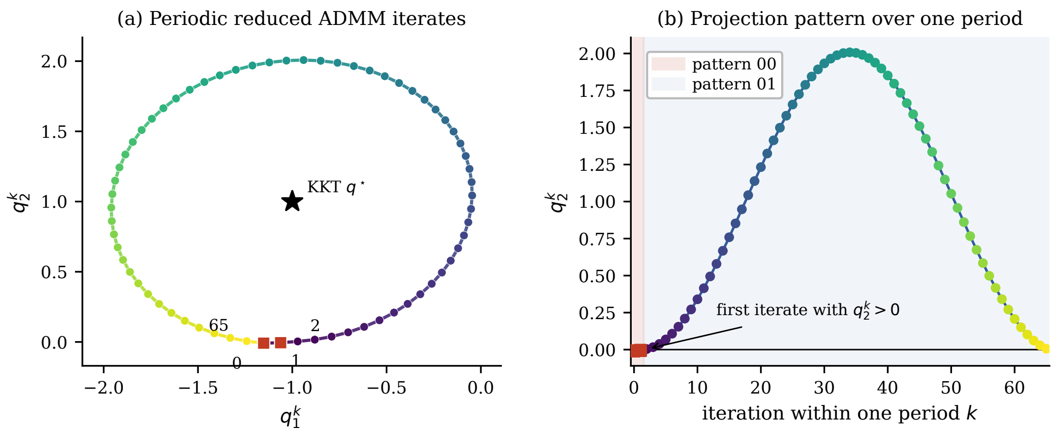

Figure 1: Exact period-66 projection pattern. Red squares mark the two 00 iterates; the remaining iterates have pattern 01. The KKT coordinate $q^\star = (-1, 1)$ is not on the periodic sequence.

2.2 Active-set geometry of the period-66 cycle

We reduce the ADMM iteration to verify the periodic sequence in Theorem 2.1.

Define the orthant-projection input

$$q^{k+1} = \bar{b} - x^{k+1} - y^{k+1} + \lambda^k. \tag{2.8}$$

The z- and multiplier updates give

$$z^{k+1} = [q^{k+1}]_+, \qquad \lambda^{k+1} = [q^{k+1}]_-, \qquad q^{k+1} = z^{k+1} + \lambda^{k+1}. \tag{2.9}$$

For $a \in \mathbb{R}^2$, write $([a]_+)_i = \max\{a_i, 0\}$, $([a]_-)_i = \min\{a_i, 0\}$, and $(|a|)_i = |a_i|$ for $i = 1, 2$. Thus (2.9) shows that the single vector q^{k+1} uniquely determines both z^{k+1} and λ^{k+1}. Since $z^{k+1} = \Pi_{\mathbb{R}^2_+}(q^{k+1})$, the projection optimality condition gives

$$\lambda^{k+1} = q^{k+1} - z^{k+1} \in N_{\mathbb{R}^2_+}(z^{k+1}).$$

We index the recurrence after the projection–multiplier update and set

$$s^k = ((y^k)^\top, (q^k)^\top)^\top \in \mathbb{R}^4.$$

Since $z - \lambda = |q|$, the eliminated variables are recovered by

$$z = [q]_+, \qquad \lambda = [q]_-, \qquad x^+ = M(\bar{b} - y - z + \lambda) = M(\bar{b} - y - |q|). \tag{2.10}$$

Define $p := \bar{b} - x^+ - z + \lambda$ and $g := (I_2 - M)\bar{b}$. Substitution gives the reduced recurrence

$$p = My - (I_2 - M)|q| + g, \tag{2.11a}$$
$$y^+ = Np, \tag{2.11b}$$
$$q^+ = \bar{b} - x^+ - y^+ + \lambda = p + z - y^+ = (I_2 - N)p + [q]_+. \tag{2.11c}$$

Together with (2.10), this recurrence is equivalent to the original ADMM after the first projection–multiplier update.

For a strict projection pattern $D = \mathrm{diag}(\mathbf{1}_{\{q_1>0\}}, \mathbf{1}_{\{q_2>0\}})$, one has $[q]_+ = Dq$ and $|q| = S_D q$, where $S_D = 2D - I_2$. Hence, on the corresponding polyhedral projection region, (2.11) is the affine update

$$\begin{aligned} s^+ &= T_D s + h, \\ T_D &= \begin{pmatrix} NM & -N(I_2 - M)S_D \\ (I_2 - N)M & D - (I_2 - N)(I_2 - M)S_D \end{pmatrix}, \quad h = \begin{pmatrix} Ng \\ (I_2 - N)g \end{pmatrix}. \end{aligned} \tag{2.12}$$

Only two projection patterns are needed: $D_{00} = \mathrm{diag}(0,0)$ and $D_{01} = \mathrm{diag}(0,1)$. Write $T_{00} := T_{D_{00}}$ and $T_{01} := T_{D_{01}}$, and represent each affine update in homogeneous coordinates by

$$L_\omega = \begin{pmatrix} T_\omega & h \\ 0 & 1 \end{pmatrix} \in \mathbb{R}^{5\times 5}, \qquad \omega \in \{00, 01\}. \tag{2.13}$$

Composing the affine updates in the order prescribed by (2.7) gives

$$L_{01}^{64} L_{00}^2 = \begin{pmatrix} \mathcal{P} & a \\ 0 & 1 \end{pmatrix}. \tag{2.14}$$

Here $\mathcal{P} \in \mathbb{Q}^{4\times 4}$ and $a \in \mathbb{Q}^4$. Exact rational elimination verifies $\det(I_4 - \mathcal{P}) \neq 0$. Hence $s^{66} = s^0$ has the unique rational solution

$$s^0 = (I_4 - \mathcal{P})^{-1} a \in \mathbb{Q}^4. \tag{2.15}$$

2.3 Why the ADMM sequence does not converge

The fixed point (2.15) defines an ADMM sequence only if its projection signs agree with $\mathcal{W}$. Exact rational iteration of (2.11) gives

$$q_1^k < 0 \quad (0 \le k < 66), \qquad q_2^k < 0 \quad (k = 0, 1), \qquad q_2^k > 0 \quad (2 \le k < 66). \tag{2.16}$$

More precisely, if $\delta_k = 0$ for $k = 0, 1$ and $\delta_k = 1$ otherwise, then

$$\min_{0 \le k < 66} \min\{-q_1^k, (2\delta_k - 1)q_2^k\} > \frac{1}{1000}. \tag{2.17}$$

Hence every branch is admissible and the orthant projection realizes $(00)^2(01)^{64}$.

Equation (2.10) recovers the full ADMM sequence. Define

$$x^0 = M(\bar{b} - y^{65} - |q^{65}|).$$

Together with $z^0 = [q^0]_+$ and $\lambda^0 = [q^0]_-$, this gives $(x^0, y^0, z^0, \lambda^0) \in \mathbb{Q}^8$, and the step-65 update returns the full state. If the ADMM sequence had a proper subperiod, its projection-sign sequence would be a proper repetition. The only possible repetition factor is two, because the sequence has length 66 and contains exactly two occurrences of 00. Its two length-33 halves, however, contain different numbers of 00 symbols. The first return therefore occurs at step 66. The periodic sequence cannot contain the unique KKT point, which is a fixed point of this single-valued ADMM iteration. It is consequently non-KKT, nonconvergent, and bounded. This proves the theorem.

3 Comparison of GPT-5.6 Sol (Codex) and Kimi Code K3

After GPT-5.6 Sol through Codex produced the rational period-66 certificate in Theorem 2.1, we gave the same problem statement, but not the Codex candidate, to Kimi Code K3 in a separate run. It followed a different computational route and produced an $m = 3$, locally attracting non-KKT sequence of minimal period 23. Appendix B gives its exact rational certificate. Because $\widehat{A}$ and $\widehat{B}$ are nonsingular, Corollary B.2 shows that an invertible blockwise change of variables converts this example to an $[I_3, I_3, I_3]$ identity-slack QP while preserving every direct-ADMM iterate.

Table 1 is an exploratory capability comparison, not a controlled benchmark. It records the realized resources, computational routes, and exact replayable certificates for the same mathematical question.

Table 1: Descriptive comparison of the two realized certificate-construction routes.

Measure	**GPT-5.6 Sol (Codex)**	**Kimi Code K3**
Approximate run time	≈ 40 hours.	≈ 9 hours.
Recorded token volume	$\approx 1.704 \times 10^9$ total; 7.869×10^6 output.	$\approx 6.327 \times 10^7$ total; 4.823×10^5 output.
Agent topology	4 top-level and 237 subagent rollouts.	One main session and 3 recorded subagents.
Certified result	Exact rational $m = 2$ identity-slack QP: bounded non-KKT sequence of minimal period 66 (Theorem 2.1).	Exact rational $m = 3$ identity-slack QP (after an invertible change of variables): a locally attracting non-KKT sequence of minimal period 23, with a certified open set of nonconvergent reduced initializations (Proposition B.1).
Computational route	Piecewise-affine reduction, structured active-set search, rationalization, and two exact replays.	Structured projector screening, KKT-compatible data construction, periodic-word detection, and exact rational certification.

Appendices B and C give the period-23 construction and detailed accounting, respectively.

4 Convergence under a dual-step perturbation

We now ask whether changing only the multiplier step can eliminate the certified cycle. The results below answer this question for the fixed rational QP (2.1) at three distinct scopes: uniform local convergence on an interval, convergence of the period-66 initialization on a smaller interval, and the exact local-stability threshold. They are not a general convergence theorem for arbitrary QPs or arbitrary initializations.

For the QP (2.1), keep the primal updates and $\beta = 1$, and replace (1.5d) by

$$\lambda^{k+1} = \lambda^k - \tau(x^{k+1} + y^{k+1} + z^{k+1} - \bar{b}), \qquad \tau > 0. \tag{4.1}$$

The standard method corresponds to $\tau = 1$. For $\tau \neq 1$, the identity $\lambda = [q]_-$ is no longer preserved, so the four-dimensional reduction in (y, q) is no longer valid. Use the six-dimensional state $w = ((y)^\top, (z)^\top, \lambda^\top)^\top \in \mathbb{R}^6$, retain $M = (Q_1 + I_2)^{-1}$ and $N = (Q_2 + I_2)^{-1}$, and write $I = I_2$. Eliminating x^+ from the primal updates, define

$$C_y = (I - N)M, \qquad C_z = N + (I - N)M, \qquad C_\lambda = (I - N)(I - M), \qquad d = C_\lambda \bar{b}.$$

The input to the orthant projection is

$$q^+ = C_y y + C_z z + C_\lambda \lambda + d.$$

For the strict target projection branch of the current sweep, set

$$D = \mathrm{diag}\big(\mathbf{1}_{\{q_1^+ > 0\}}, \mathbf{1}_{\{q_2^+ > 0\}}\big), \qquad z^+ = Dq^+.$$

Since $x^+ + y^+ + z^+ - \bar{b} = \lambda - q^+ + z^+$, the relaxed multiplier update is

$$\lambda^+ = (1 - \tau)\lambda + \tau(I - D)q^+.$$

Consequently, the original four updates restrict on this branch to

$$w^+ = T_D(\tau)w + a_D(\tau), \qquad T_D(\tau) = A_D + \tau E_D, \tag{4.2}$$

where the 2×2 block matrices are

$$A_D = \begin{pmatrix} NM & -N(I - M) & N(I - M) \\ DC_y & DC_z & DC_\lambda \\ 0 & 0 & I \end{pmatrix},$$

$$E_D = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ (I - D)C_y & (I - D)C_z & (I - D)C_\lambda - I \end{pmatrix},$$

and the affine term is

$$a_D(\tau) = \begin{pmatrix} N(I - M)\bar{b} \\ Dd \\ 0 \end{pmatrix} + \tau \begin{pmatrix} 0 \\ 0 \\ (I - D)d \end{pmatrix}.$$

Thus τ enters only the multiplier block row; the x-, y-, and z-updates are unchanged.

For the counterexample, $q^\star = (-1, 1)$ and $D_\star = D_{01}$. Figure 2 summarizes the three scopes below.

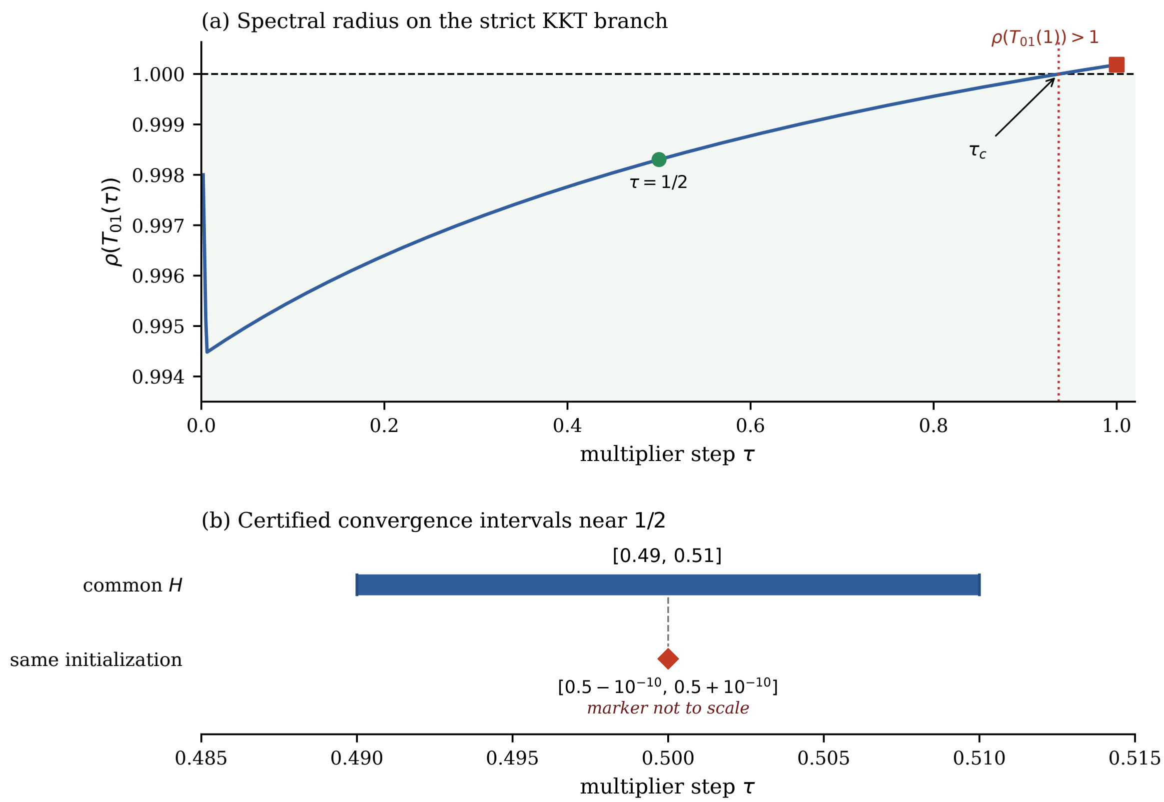

Figure 2: Certified dual-step ranges for the fixed rational instance: pointwise local attraction on the strict D_{01} branch (equivalently, Schur stability), a uniform common-Lyapunov interval, and convergence of the initialization in Theorem 2.1.

Theorem 4.1 (Dual step length for the period-66 instance)**.** *For the rational QP* (2.1) *and the relaxed multiplier update* (4.1)*:*

1. *one rational Lyapunov matrix and one projection-safe neighborhood give uniform local linear convergence to the unique KKT point for every*

$$\frac{49}{100} \leq \tau \leq \frac{51}{100};$$

2. *the period-*66 *initialization of Theorem 2.1 converges to that KKT point for every*

$$\frac{4999999999}{10^{10}} \leq \tau \leq \frac{5000000001}{10^{10}};$$

3. *for each fixed $0 < \tau < 1$, the unique KKT point is a locally linearly attracting fixed point of the full projected iteration if and only if $0 < \tau < \tau_c$. Equivalently, the local error matrix $T_{01}(\tau)$ is Schur stable precisely on this interval, where*

$$0.9366061114 < \tau_c < 0.9366061115.$$

The value τ_c is the unique root in $(0, 1)$ of the polynomial G in (A.1)*. The neighborhood and contraction metric in this third assertion may depend on τ.*

Computer-assisted proof. Let $T(\tau) = T_{01}(\tau) = A + \tau E$ and $\Phi_H(\tau) = H - T(\tau)^\top H T(\tau)$. If $\tau = \eta\tau_- + (1-\eta)\tau_+$, direct expansion gives

$$\Phi_H(\tau) - \eta\Phi_H(\tau_-) - (1-\eta)\Phi_H(\tau_+) = \eta(1-\eta)(\tau_+ - \tau_-)^2 E^\top H E \succeq 0. \tag{4.3}$$

At $\tau = 1/2$, let $H \in \mathbb{Q}^{6\times 6}$ solve

$$H - T_{01}(1/2)^\top H T_{01}(1/2) = I_6.$$

Exact Sylvester tests give $H \succ 0$, $\Phi_H(49/100) \succ 0$, and $\Phi_H(51/100) \succ 0$. Equation (4.3) yields uniform contraction on the interval; strict complementarity gives a projection-invariant H-ellipsoid around the KKT point. This proves the first claim.

For the initialization in Theorem 2.1, a 232-step rational componentwise enclosure is propagated uniformly for $|\tau - 1/2| \le 10^{-10}$. Every projection sign stays strict, and at step 232 the entire enclosure lies inside the common projection-safe Lyapunov ellipsoid; the first claim then gives convergence. Finally, exact factorization of the D_{01}-region characteristic polynomial, followed by a real Schur recursion and a Sturm root count, reduces the stability boundary to the unique root in $(0,1)$ of G in (A.1). Rational endpoint evaluations give the displayed bracket.

It remains to connect this branch spectrum to the full projected iteration. Fix $0 < \tau < \tau_c$. Schur stability gives a unique $H_\tau \succ 0$ satisfying

$$H_\tau - T_{01}(\tau)^\top H_\tau T_{01}(\tau) = I_6.$$

With $\|e\|_{H_\tau}^2 = e^\top H_\tau e$, there is a $\kappa_\tau < 1$ such that

$$\|T_{01}(\tau)e\|_{H_\tau} \le \kappa_\tau \|e\|_{H_\tau}.$$

Because $q^\star = (-1,1)$ lies strictly inside the D_{01} projection region, a sufficiently small closed H_τ-ellipsoid $\mathcal{E}_\tau$ around $w^\star$ lies entirely in that region. On $\mathcal{E}_\tau$, the full projected ADMM map equals its D_{01} affine branch and maps $\mathcal{E}_\tau$ into itself. It is therefore a contraction on the complete metric space $\mathcal{E}_\tau$; the Banach fixed-point theorem gives the unique fixed point $w^\star$ and linear convergence to it from every point of $\mathcal{E}_\tau$. Conversely, for $\tau_c \le \tau < 1$, the branch matrix has spectral radius at least one. Since the KKT point is in the interior of the same affine branch, it cannot be locally linearly attracting. This proves the third claim. Appendix A gives the verification commands. □

The two convergence intervals are sufficient and are not claimed to be maximal; the theorem makes no global claim over arbitrary initializations.

5 Conclusion

The identity third constraint block does not guarantee global convergence of direct three-block ADMM. Theorem 2.1 gives a two-dimensional strongly convex rational QP with a bounded non-KKT sequence of minimal period 66. The Kimi Code K3 result in Proposition B.1, stated in Appendix B, gives a three-dimensional instance with a locally attracting non-KKT sequence of period 23, and hence an open set of nonconvergent reduced initializations.

For the period-66 instance, modifying only the dual step length yields local linear convergence on a certified interval and convergence of the initialization in Theorem 2.1 on a smaller finite-entry interval. These results are local or initialization-specific. Determining sharp dual-step ranges and structural conditions for global convergence remains open.

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A Certificates for the period-66 and dual-step results

A.1 Exact replay for Theorem 2.1

From the period-66 certificate repository root, run

```
python python/verify_certificate_pair.py.
```

The command invokes separately implemented four- and six-dimensional rational replays and verifies strong convexity, the unique KKT point, every ADMM update, 66-step closure, all 132 strict projection inequalities, and minimality of the period.

A.2 Persistence under perturbations of the problem data

Corollary A.1 (Persistence under data perturbations)**.** *Let $\mathbb{S}^2_{++}$ denote the cone of real 2×2 symmetric positive-definite matrices. With $A = B = I_2$, $\beta = 1$, and the update order $x \to y \to z \to \lambda$ fixed, there is an open neighborhood of $(Q_1, Q_2, \bar{b})$ in $\mathbb{S}^2_{++} \times \mathbb{S}^2_{++} \times \mathbb{R}^2$ such that each resulting problem has a non-KKT periodic ADMM sequence of minimal period 66, with the same projection-sign sequence and every projection inequality strict.*

Proof. For the fixed projection-sign sequence, $\mathcal{P}$ and a in (2.14) depend analytically on the problem data. The period system remains nonsingular near the displayed instance, so $(I_4 - \mathcal{P})^{-1}a$ and its 66 reduced iterates vary continuously. The signs in (2.17) have a positive common margin and remain strict under sufficiently small perturbations. The primitive word is unchanged. $\square$

A.3 Verification of Theorem 4.1

The dual-step certificate is verified by

```
python python/certify_relaxed_multiplier_interval_theory.py.
```

It re-derives the full-state projection-region matrices from the original updates, checks the endpoint Sylvester minors and chord identity, propagates the exact 232-step finite-entry enclosure, and performs the Schur–Sturm boundary test. The boundary polynomial is

$$\begin{aligned} G(\tau) = {}& 111794210406295556649228900462157733493\,\tau^3 \\ & + 2310577697528181610827581444128442208517 1521\,\tau^2 \\ & - 2441573397158988214402436496739594630715 43521\,\tau \\ & + 20841006066046034038657663888981457882863 8507. \end{aligned} \tag{A.1}$$

An exact Sturm count gives one root of G in $(0, 1)$, and rational endpoint evaluation gives the bracket in Theorem 4.1.

B Exact certificate for the period-23 counterexample

This appendix records the exact $m = 3$ certificate obtained by the Kimi Code K3 route. It is a fixed-QP initialization result and does not assert robustness to perturbations of the problem data.

B.1 Reduced branch map

Record the state immediately after the z- and multiplier updates. With $\beta = 1$, set

$$t = z + \lambda, \qquad v = (y, t) \in \mathbb{R}^6, \qquad S_t = [\, 0_{3\times3} \; I_3 \,],$$

so that $t = S_t v$. The projection identity gives $z = [t]_+$ and $\lambda = [t]_-$; thus v determines (z, λ) and the next ADMM step.

Each strict projection branch is determined by

$$\sigma(t) := (\operatorname{sgn} t_1, \operatorname{sgn} t_2, \operatorname{sgn} t_3) \in \{-,+\}^3.$$

On a fixed branch the projection is linear, and eliminating the x- and z-updates gives

$$\Phi_\sigma(v) = R_\sigma v + r_\sigma, \qquad R_\sigma \in \mathbb{Q}^{6\times 6}, \quad r_\sigma \in \mathbb{Q}^6.$$

Set

$$K_x = (F + A^\top A)^{-1}, \quad K_y = (G + B^\top B)^{-1}, \qquad H_x = AK_xA^\top, \quad H_y = BK_yB^\top,$$
$$D_\sigma = \operatorname{diag}\big(\mathbf{1}_{\{\sigma_i=+\}}\big), \quad J_\sigma = 2D_\sigma - I_3, \quad E_\sigma = I_3 - D_\sigma,$$
$$\xi = K_x(A^\top b - c_1), \qquad \eta = K_y(B^\top b - c_2 - B^\top A\xi).$$

Direct substitution in the x-, y-, and projection–multiplier updates gives

$$R_\sigma = \begin{bmatrix} K_yB^\top H_xB & K_yB^\top(H_x - I_3)J_\sigma \\ (I_3 - H_y)H_xB & E_\sigma + \big((I_3 - H_y)H_x + H_y\big)J_\sigma \end{bmatrix}, \qquad r_\sigma = \begin{bmatrix} \eta \\ b - A\xi - B\eta \end{bmatrix}. \tag{B.1}$$

If the QP data are rational, then every entry of R_σ and r_σ is rational.

B.2 Rational counterexample and exact certificate

With $\beta = 1$, apply the reduced map above to the following rational instance:

$$\widehat{A} = \begin{bmatrix} 5/36 & 1/60 & 25/97 \\ 1/60 & 5/77 & 5/91 \\ 25/97 & 5/91 & 11/12 \end{bmatrix}, \qquad \widehat{B} = \begin{bmatrix} 4/61 & 1/54 & 5/83 \\ 1/54 & 13/92 & 23/88 \\ 5/83 & 23/88 & 53/58 \end{bmatrix},$$
$$\widehat{F} = \begin{bmatrix} 85/93 & -1/57 & -3/11 \\ -1/57 & 99/100 & -5/86 \\ -3/11 & -5/86 & 7/78 \end{bmatrix}, \qquad \widehat{G} = \begin{bmatrix} 99/100 & -1/51 & -3/47 \\ -1/51 & 72/79 & -18/65 \\ -3/47 & -18/65 & 4/43 \end{bmatrix}, \tag{B.2}$$
$$\widehat{b} = \begin{bmatrix} -2/17 \\ -1/14 \\ 56/73 \end{bmatrix}, \qquad \widehat{c}_1 = \begin{bmatrix} 33/98 \\ -33/25 \\ -24/55 \end{bmatrix}, \qquad \widehat{c}_2 = \begin{bmatrix} -31/99 \\ 19/36 \\ -11/20 \end{bmatrix}.$$

Every displayed fraction p/q is in lowest terms and satisfies $|p| \le 100$ and $1 \le q \le 100$. The resulting strict projection-sign word is

$$\mathcal{W}_{23} = (+,-,+)^5\,(-,+,+)^7\,(-,-,+)^2\,(-,-,-)\,(-,-,+)^8. \tag{B.3}$$

Take $\omega = \mathcal{W}_{23} = (\sigma_0, \dots, \sigma_{22})$. Write the composition of its first j branch maps as

$$\Phi_\omega^{(j)}(v) = L_jv + d_j, \qquad L_0 = I_6, \quad d_0 = 0,$$
$$L_{j+1} = R_{\sigma_j}L_j, \qquad d_{j+1} = R_{\sigma_j}d_j + r_{\sigma_j}.$$

The full 23-step return map is therefore

$$\begin{aligned} v^{23(\ell+1)} &= \Phi_{\text{per}}\big(v^{23\ell}\big), \\ \Phi_{\text{per}}(v) &:= \Phi_\omega^{(23)}(v) = M_{\text{per}}v + c_{\text{per}}, \\ M_{\text{per}} &:= L_{23} = R_{\sigma_{22}} \cdots R_{\sigma_0}, \qquad c_{\text{per}} := d_{23}. \end{aligned} \tag{B.4}$$

Exact elimination shows that $I_6 - M_{\mathrm{per}}$ is nonsingular, so define the phase-zero state by

$$\widehat{v}^0 := (I_6 - M_{\mathrm{per}})^{-1} c_{\mathrm{per}} \in \mathbb{Q}^6, \qquad \widehat{v}^j := L_j \widehat{v}^0 + d_j \quad (j = 0, \dots, 22).$$

The complete exact rational initialization is recovered by

$$\widehat{t}^0 = S_t \widehat{v}^0, \qquad \widehat{z}^0 = [\widehat{t}^0]_+, \qquad \widehat{\lambda}^0 = [\widehat{t}^0]_-,$$
$$\widehat{K}_x := (\widehat{F} + \widehat{A}^\top \widehat{A})^{-1}, \qquad \widehat{x}^0 = \widehat{K}_x \Big[\widehat{A}^\top (\widehat{b} - \widehat{B} \widehat{y}^{22} - \widehat{z}^{22} + \widehat{\lambda}^{22}) - \widehat{c}_1 \Big].$$

These equations define the exact rational initialization; the machine certificate stores its expanded numerators and denominators.

Proposition B.1 (Exact period-23 certificate for nearby initializations)**.** *For the rational data* (B.2)*, $\widehat{F}, \widehat{G} \succ 0$ and $\det(\widehat{A}) \det(\widehat{B}) \neq 0$. The direct iteration* (1.5)*, with $\beta = 1$, applied to*

$$\begin{aligned} \min_{x,y,z \in \mathbb{R}^3} \quad & \frac{1}{2} x^\top \widehat{F} x + \widehat{c}_1^\top x + \frac{1}{2} y^\top \widehat{G} y + \widehat{c}_2^\top y + \iota_{\mathbb{R}^3_+}(z) \\ \text{s.t.} \quad & \widehat{A} x + \widehat{B} y + z = \widehat{b}, \end{aligned} \tag{B.5}$$

has the following properties.

(i) *It has a unique primal–dual KKT point.*

(ii) *Direct ADMM has a non-KKT sequence of minimal period* 23*. Its strict projection-sign word is* (B.3)*; its exponents denote consecutive repetitions and sum to* 23*. All* $23 \times 3 = 69$ *projection inputs are separated from zero:*

$$\min_{0 \le k < 23} \min_{1 \le i \le 3} |\widehat{t}_i^k| > \frac{1}{250}.$$

(iii) *There is a rational matrix $P \succ 0$ for which*

$$\mathcal{E}_{\mathrm{cert}} := \left\{ \widehat{v}^0 + e : e^\top P e < \frac{1}{4000} \right\} \tag{B.6}$$

is an open invariant set for the 23*-step return map* (B.4)*. Every $v^0 \in \mathcal{E}_{\mathrm{cert}}$ follows the repeating pattern $\mathcal{W}_{23}$ and converges phasewise to the period-*23 *sequence.*

Consequently, the reconstructed ADMM sequence is nonconvergent for every reduced initialization $v^0 \in \mathcal{E}_{\mathrm{cert}}$.

Proof. For $\omega = \mathcal{W}_{23}$, the recursion above gives the exact rational return map

$$(M_{\mathrm{per}}, c_{\mathrm{per}}) = (L_{23}, d_{23}).$$

Exact rational elimination solves the fixed-point system for $\widehat{v}^0$. Exact replay reconstructs every phase state and verifies the pattern, the strict margin, $\Phi_{\mathrm{per}}(\widehat{v}^0) = \widehat{v}^0$, and that the 23 phase states are pairwise distinct. Hence the sequence has minimal period 23.

Positive definiteness of $\widehat{F}$ and $\widehat{G}$ gives a unique primal solution; feasibility determines z, and nonsingularity of $\widehat{A}$ determines λ. Thus the KKT point is unique and is a fixed point of the single-valued ADMM map. It cannot belong to the nonconstant period-23 sequence, which is therefore non-KKT.

For nearby initializations, the matrix P in Appendix B.3 satisfies

$$P - M_{\mathrm{per}}^{\top} P M_{\mathrm{per}} \succ 0.$$

It follows that the error decreases after each complete block of 23 ADMM steps. In particular, every eigenvalue of M_{per} lies strictly inside the unit disk. The strict sign margin guarantees that every point in $\mathcal{E}_{\mathrm{cert}}$ continues to use the same 23 affine branches. The return map sends this ellipsoid into itself, so induction gives the same projection pattern and phasewise convergence for all subsequent returns. The 23 limiting phase points are distinct; hence the full sequence does not converge to a single point. $\square$

Corollary B.2 (Identity-slack equivalence)**.** *Set*

$$u = \widehat{A}x, \qquad w = \widehat{B}y.$$

Then problem (B.5) *is equivalent to*

$$\begin{aligned} \min_{u,w,z\in\mathbb{R}^3} \quad & \frac{1}{2}u^\top \widetilde{F}u + \widetilde{c}_1^\top u + \frac{1}{2}w^\top \widetilde{G}w + \widetilde{c}_2^\top w + \iota_{\mathbb{R}^3_+}(z) \\ s.t. \quad & u + w + z = \widehat{b}, \end{aligned}$$

where

$$\widetilde{F} = \widehat{A}^{-\top}\widehat{F}\widehat{A}^{-1}, \quad \widetilde{G} = \widehat{B}^{-\top}\widehat{G}\widehat{B}^{-1}, \quad \widetilde{c}_1 = \widehat{A}^{-\top}\widehat{c}_1, \quad \widetilde{c}_2 = \widehat{B}^{-\top}\widehat{c}_2.$$

The transformed data remain rational and $\widetilde{F}, \widetilde{G} \succ 0$*. Thus Proposition B.1 also gives an exact rational, locally attracting period-*23 *counterexample for the* $[I_3, I_3, I_3]$ *identity-slack model.*

Proof. Nonsingularity of $\widehat{A}$ and $\widehat{B}$ makes the change of variables bijective and preserves the constraint residual:

$$\widehat{A}x + \widehat{B}y + z - \widehat{b} = u + w + z - \widehat{b}.$$

The x- and y-subproblems are therefore invertible reparameterizations of the u- and w-subproblems, while the z- and multiplier updates are unchanged. Starting from $u^0 = \widehat{A}x^0$ and $w^0 = \widehat{B}y^0$, the two direct ADMM trajectories consequently correspond step by step. Positive definiteness is preserved by congruence, and inverses of nonsingular rational matrices remain rational. $\square$

B.3 Return-invariant neighborhood

Use the partial-composition matrices L_j and the selector S_t defined in Appendix B.1. Set

$$P = \frac{1}{2}\begin{bmatrix} 2 & 0 & 0 & 0 & -1 & 1 \\ 0 & 2 & 1 & 0 & -2 & 2 \\ 0 & 1 & 6 & 2 & -7 & 7 \\ 0 & 0 & 2 & 3 & -3 & 3 \\ -1 & -2 & -7 & -3 & 16 & -13 \\ 1 & 2 & 7 & 3 & -13 & 14 \end{bmatrix}.$$

Its leading principal minors are

$$1, \quad 1, \quad \frac{11}{4}, \quad \frac{25}{8}, \quad \frac{333}{32}, \quad \frac{441}{32},$$

so $P \succ 0$. Exact Sylvester tests also give

$$P - M_{\mathrm{per}}^{\top} P M_{\mathrm{per}} \succ 0.$$

For the ith standard basis vector $\mathbf{e}_i \in \mathbb{R}^3$, define

$$a_{ji} := \mathbf{e}_i^{\top} S_t L_j \in \mathbb{Q}^{1\times 6}.$$

Thus $a_{ji}e$ is the perturbation of the ith projection input at phase j caused by the initial-state error $e = v^0 - \widehat{v}^0$. Define the corresponding sign-preserving radius by

$$\bar{r}^2 := \min_{\substack{0\le j<23,\ 1\le i\le 3\\ a_{ji}\neq 0}} \frac{(\widehat{t}_i^{\,j})^2}{a_{ji}P^{-1}a_{ji}^{\top}}.$$

The exact certificate gives $\bar{r}^2 > 29/100000 > 1/4000$; the minimum occurs at phase 14 and coordinate t_3. The P-norm Cauchy–Schwarz inequality therefore preserves all 69 projection signs for $e^{\top}Pe < 1/4000$. Since $P - M_{\mathrm{per}}^{\top} P M_{\mathrm{per}} \succ 0$, the return map leaves $\mathcal{E}_{\mathrm{cert}}$ invariant. Induction gives the same branch word and phasewise convergence to the non-KKT sequence.

The ellipsoid in (B.6) is a sufficient initialization set; it is not claimed to be the largest one. On the slice $\Delta y = 0$, $\Delta t_3 = 0$, its quadratic form is

$$\frac{3}{2}(\Delta t_1)^2 - 3\Delta t_1 \Delta t_2 + 8(\Delta t_2)^2 < \frac{1}{4000}.$$

Setting $\Delta t_2 = 0$ gives the explicit interval

$$|\Delta t_1| < \frac{1}{\sqrt{6000}}.$$

Every point in this interval is therefore a certified nonconvergent reduced initialization.

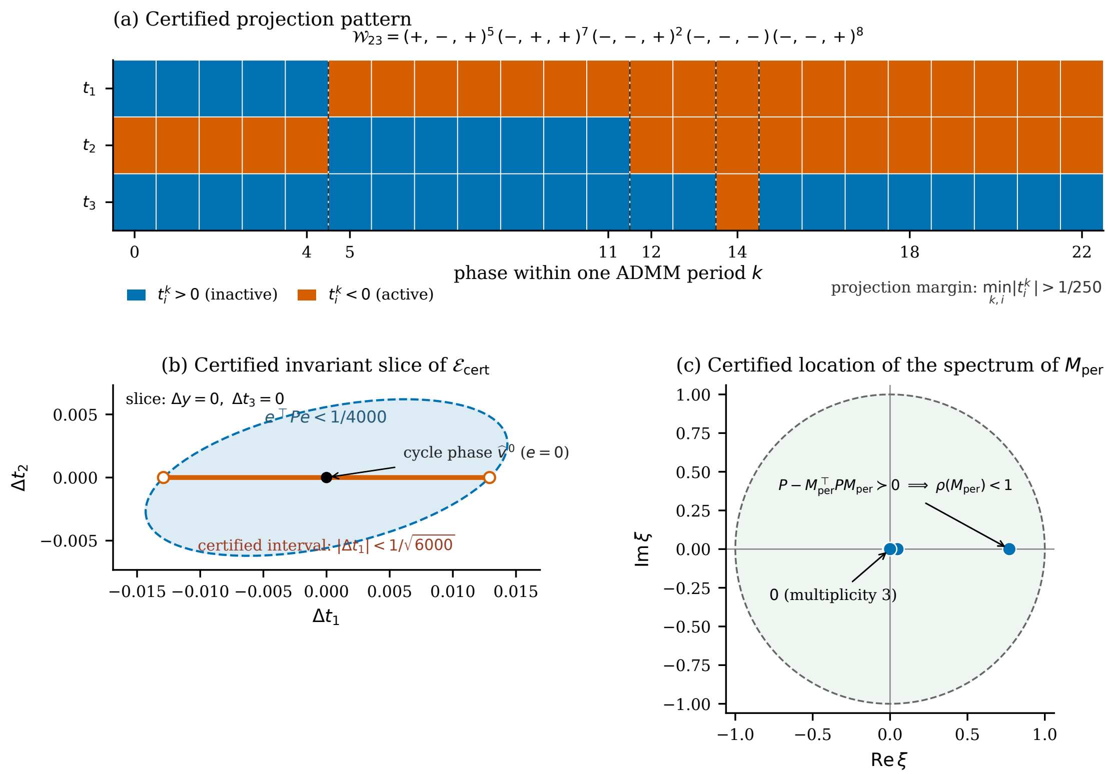

Figure 3: Exact certificate for the locally attracting period-23 sequence. (a) The certified projection word $\mathcal{W}_{23}$: blue denotes $t_i > 0$ (inactive inequality), vermillion denotes $t_i < 0$ (active inequality), and every projection input has margin $> 1/250$. (b) A two-dimensional slice of $\mathcal{E}_{\mathrm{cert}}$. The dashed ellipse is a sufficient invariant neighborhood, and the vermillion segment is the explicit certified interval $|\Delta t_1| < 1/\sqrt{6000}$ at $\Delta t_2 = 0$. (c) Location of the spectrum of M_{per}. The exact Lyapunov inequality $P - M_{\mathrm{per}}^\top P M_{\mathrm{per}} \succ 0$ certifies $\rho(M_{\mathrm{per}}) < 1$. No claim is made outside the certified ellipse.

B.4 Reproduction and source data

From the public certificate repository root, running

```
python python/verify_period23_certificate.py
```

verifies positive definiteness, nonsingularity, the unique KKT point, 23-step closure, 23 distinct phase states, strict projection consistency with margin $> 1/250$, separation from the KKT point, and the exact Lyapunov inequality for the return matrix. The rational input, verifier, and generated certificate are provided in the accompanying repository.

C Computational provenance and retained artifacts

The comparison in Section 3 is descriptive, not a controlled model benchmark. The Codex workflow included human interventions in problem specification, evidence requirements, and search priorities. The Kimi workflow began from a frozen workspace without Codex candidates, but was audited post hoc rather than preregistered.

The first exact replayable nonconvergence certificate defines the time, token, and topology endpoint. De-duplicated completed-turn intervals give active wall time; their sums give cumulative agent-time, 75.06 hours for Codex and 12.43 hours for Kimi. The 36–44-hour Codex estimate removes estimated setup and restart overhead from the observed 52.14 hours; it is not an observed rerun. Token totals include cached input, noncached input, and output. Because telemetry and tokenizers differ, they measure recorded traffic rather than billable usage.

C.1 Retained artifacts

The theoretical inventory includes not only the terminal counterexamples but also reductions, conditional convergence theorems, and restricted exclusions closed during the search. The table separates rigorous results from mechanism explanations, search outputs, and verification files. These artifacts do not define the timing endpoint used in Table 1.

Asset	GPT-5.6 Sol (Codex)	Kimi Code K3
Reductions and structure	Projection identity; signed piecewise-affine, source–target, and effective-state reductions; phase matrices and affine period equations.	The reduction $t = z + \lambda$; piecewise-affine mode matrices on sign cones; the affine return map for a prescribed projection word.
Closed theoretical results	The nonconstant two-dimensional length-2 switching-product theorem; all-parameter scalar convergence; fixed-QP phase-Lyapunov convergence; and parameter-box, small-gain, Selector-IQC, and arbitrary-dimensional common-metric sufficient conditions.	Theorem S (real eigenvalues at most 1); Theorem M1 (global convergence for the homogeneous quadratic class with $m = 1$); and Propositions C1–C3 (spectral criteria for mode stabilization, local repulsion, and local convergence).
Exclusions and mechanisms	Exact exclusions of real expansion in a fixed region, several short projection words, and selected nested words of length two through four; restricted obstructions for the direct VI/PPA condition, one-step KKT energy, and diagonal phase metrics. Each has an explicit scope.	The real-mode obstruction excludes real expansion within one cone. Complex-mode repulsion, cross-cone switching, and frequency locking explain cycle formation; the latter two are mechanisms, not general convergence or nonconvergence theorems.
Terminal counterexample	An exact rational $m = 2$ identity-slack QP; a bounded non-KKT sequence of minimal period 66; 132 strict projection signs; and separate exact replays in four and six dimensions.	An exact rational $m = 3$ QP; a locally attracting non-KKT sequence of minimal period 23; 69 strict projection signs; and exact replay plus Lyapunov/Jury stability certificates.
Search artifacts	9 top-level scenario classes. Nested screens included 87,348 length-2–8 projection-sign sequences, 157 canonical margin patterns, and 340 exact tagged patterns for one fixed rational QP.	Scans covered 30,000 instances and 742,752 modes, then 40,000 projector instances with 153 expansion hits; 45 final settings produced 8 saved attracting sequences.
Verification artifacts	Separately implemented four- and six-dimensional exact replays, precision and MATLAB checks; 261 collectable pytest items, 68 certificate JSON files, and 65 review packages.	26 experiment/test scripts, 18 JSON files, and 9 NPZ files; the formal certificate package contains an exact rational replay and Lyapunov check.